

PSEUDOCODE DAN FLOWCHART KELOMPOK G

a. Pseudocode

1. Program 1 :

```
PROGRAM Hitung_Kalimat_Dan_Kata

DEKLARASI
    teks          : STRING
    JumlahKalimat : INTEGER
    JumlahKata     : INTEGER

ALGORITMA
    READ(teks)

    JumlahKalimat ← HITUNG_KARAKTER(teks, '.')
    JumlahKata    ← HITUNG_ELEMEN(PISAHKAN(teks, ' '))

    WRITE("Teks tersebut memuat ",
          JumlahKalimat,
          " kalimat dan ",
          JumlahKata,
          " kata.")

END PROGRAM
```

2. Program 2 :

```
PROGRAM String_Kalimat

DEKLARASI
    K1 : STRING
    K2 : STRING
    K3 : STRING
    K4 : STRING

ALGORITMA
    K1 ← "Saya tak 'kan menyerah."
    K2 ← 'Ia berkata, "Aku menyayangimu."'
    K3 ← "\"Coba jelaskan pengertian 'cross-validation' dalam  
Machine Learning!\""
    K4 ← "Surat keputusan itu bernomor 62/UN.34/19/2023."

    WRITE(K1)
    WRITE(K2)
    WRITE(K3)
    WRITE(K4)

END PROGRAM
```

3. Program 3 :

```
PROGRAM Hitung_Akar_Persamaan_Kuadrat
```

```
DEKLARASI
```

```
  a, b, c      : REAL  
  D            : REAL  
  x, x1, x2    : REAL
```

```
ALGORITMA
```

```
  READ(a, b, c)
```

```
  D ← (b * b) - (4 * a * c)
```

```
  IF D = 0 THEN
```

```
    x ← -b / (2 * a)
```

```
    WRITE("Nilai x kembar = ", FORMAT_DESIMAL(x, 3))
```

```
  ELSE
```

```
    IF D > 0 THEN
```

```
      x1 ← (-b + AKAR(D)) / (2 * a)
```

```
      x2 ← (-b - AKAR(D)) / (2 * a)
```

```
      WRITE("Akar real (3 desimal):")
```

```
      WRITE("x1 = ", FORMAT_DESIMAL(x1, 3))
```

```
      WRITE("x2 = ", FORMAT_DESIMAL(x2, 3))
```

```
    ELSE
```

```
      WRITE("Persamaan hanya memiliki akar imajiner")
```

```
    ENDIF
```

```
  ENDIF
```

```
END PROGRAM
```

4. Program 4 :

```
PROGRAM NIP_ASN
```

```
DEKLARASI
```

```
  NIP          : STRING  
  tahun        : INTEGER  
  bulan        : INTEGER  
  tanggal     : INTEGER  
  nama_bulan   : STRING  
  max_hari     : INTEGER
```

```
  data_bulan   : ARRAY
```

```
  data_hari    : ARRAY
```

```
ALGORITMA
```

```

data_bulan ← [1:"Januari", 2:"Februari", 3:"Maret",
              4:"April", 5:"Mei", 6:"Juni",
              7:"Juli", 8:"Agustus", 9:"September",
              10:"Oktober", 11:"November", 12:"Desember"]

data_hari ← [1:31, 2:28, 3:31, 4:30, 5:31, 6:30,
            7:31, 8:31, 9:30, 10:31, 11:30, 12:31]

LABEL_INPUT:
  READ(NIP)

  tahun ← INT(NIP[1:4])
  bulan ← INT(NIP[5:6])
  tanggal ← INT(NIP[7:8])

  IF (bulan < 1) OR (bulan > 12) THEN
    WRITE("NIP tidak valid, harap masukkan ulang.")
    GOTO LABEL_INPUT
  ENDIF

  nama_bulan ← data_bulan[bulan]
  max_hari ← data_hari[bulan]

  IF (bulan = 2) AND
    (tahun MOD 400 = 0 OR
    (tahun MOD 4 = 0 AND tahun MOD 100 ≠ 0)) THEN
    max_hari ← 29
  ENDIF

  IF (tanggal < 1) OR (tanggal > max_hari) THEN
    WRITE("NIP tidak valid, harap masukkan ulang.")
    GOTO LABEL_INPUT
  ENDIF

  WRITE("Tanggal lahir Anda: ",
        tanggal, " ",
        nama_bulan, " ",
        tahun)

END PROGRAM

```

5. Program 5 :

```

PROGRAM KlasifikasiCluster3D
DEKLARASI
  x1, x2, x3 : REAL
  dA, dB, dC : REAL
ALGORITMA
  WRITE("=== Sistem Klasifikasi Cluster 3D ===")
  WRITE("Titik Pusat Cluster Tetap:")
  WRITE("A(2,1,3), B(1,-4,6), C(-2,3,-2)")

```

```

WRITE("Masukkan koordinat U (x1, x2, x3):")
READ(x1, x2, x3)

IF x1 BUKAN_ANGKA OR
  x2 BUKAN_ANGKA OR
  x3 BUKAN_ANGKA THEN

  WRITE("Tampilkan Salah Input")

ELSE

  dA ← SQRT((x1 - 2)^2 + (x2 - 1)^2 + (x3 - 3)^2)
  dB ← SQRT((x1 - 1)^2 + (x2 - (-4))^2 + (x3 - 6)^2)
  dC ← SQRT((x1 - (-2))^2 + (x2 - 3)^2 + (x3 - (-2))^2)

  IF (dA = dB) < dC OR (dB = dC) < dA OR (dA = dC) < dB THEN

    WRITE("Tampilkan Kondisi Khusus / Nilai Jarak Sama")

  ELSE IF (dA < dB AND dB < dC) OR (dA < dC AND dC < dB) THEN

    WRITE("Tampilkan Cluster A dan Nilai Jarak")

  ELSE IF (dB < dC AND dC < dA) OR (dB < dA AND dA < dC) THEN

    WRITE("Tampilkan Cluster B dan Nilai Jarak")

  ELSE

    WRITE("Tampilkan Cluster C dan Nilai Jarak")

  ENDIF

ENDIF

ENDIF

END PROGRAM

```

6. Program 6 :

Program Menghitung Interval Konfidensi

DEKLARASI

p_hat, alpha, z : REAL

margin_of_error, batas_bawah, batas_atas : REAL

```

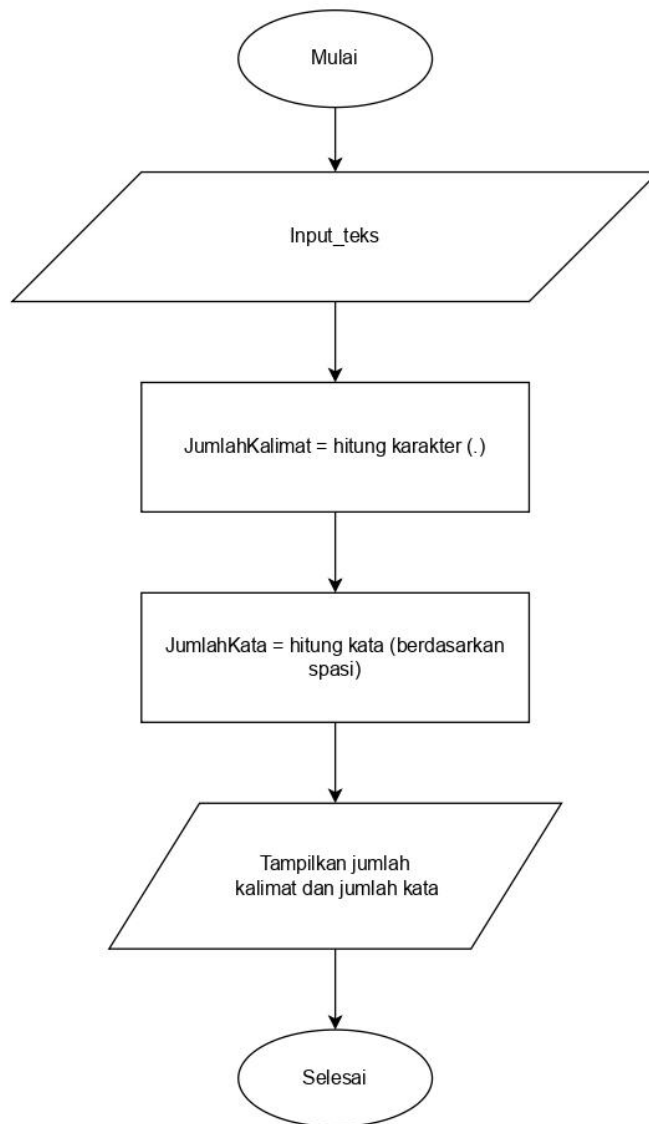
n : INTEGER

BEGIN
    Input p_hat, alpha, n
    //Validasi input proporsi
    IF (p_hat < 0 OR p_hat > 1) THEN
        PRINT("Input harus bernilai antara 0 dan 1")
        EXIT
    ENDIF
    //Menentukan nilai z berdasarkan alpha
    IF(alpha == 0.1) THEN
        z <- 1.645
    ELSE IF (alpha == 0.05) THEN
        z <- 1.96
    ELSE
        PRINT("Nilai alpha hanya mendukung 0.05 dan 0.1")
        EXIT
    ENDIF
    //Pehitungan
    margin_of_error <- z * SQRT((p_hat * (1-p_hat))/n)
    batas_bawah <- p_hat - margin_of_error
    batas_atas <- p_hat + margin_of_error
    //Output
    Print("Interval konfidensi :", batas_bawah, "< p <",
    batas_atas)
END PROGRAM

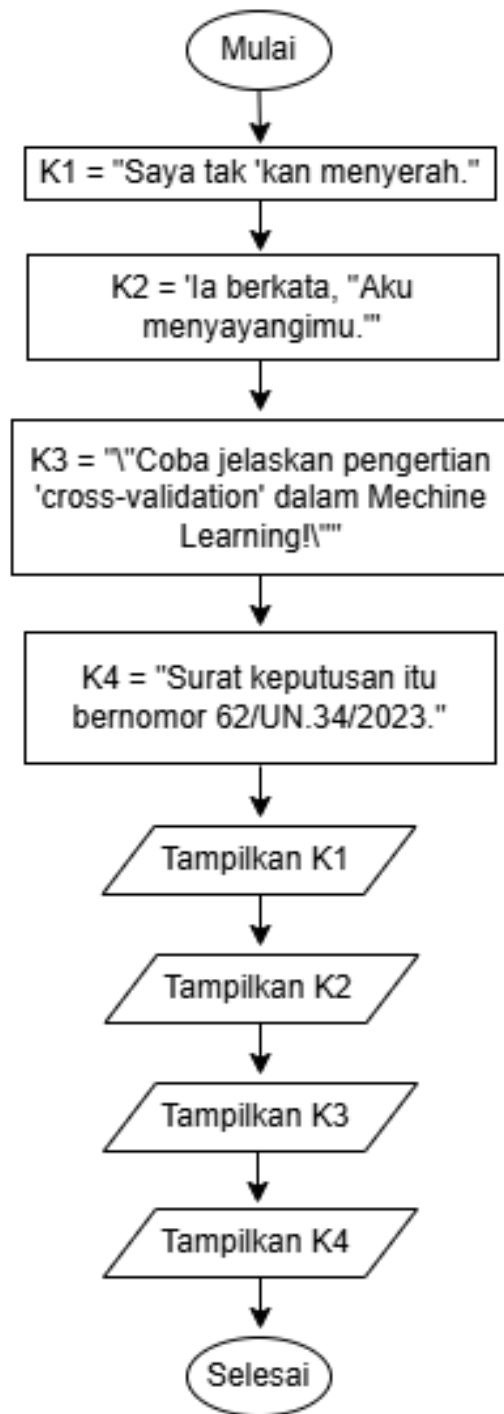
```

b. Flowchart

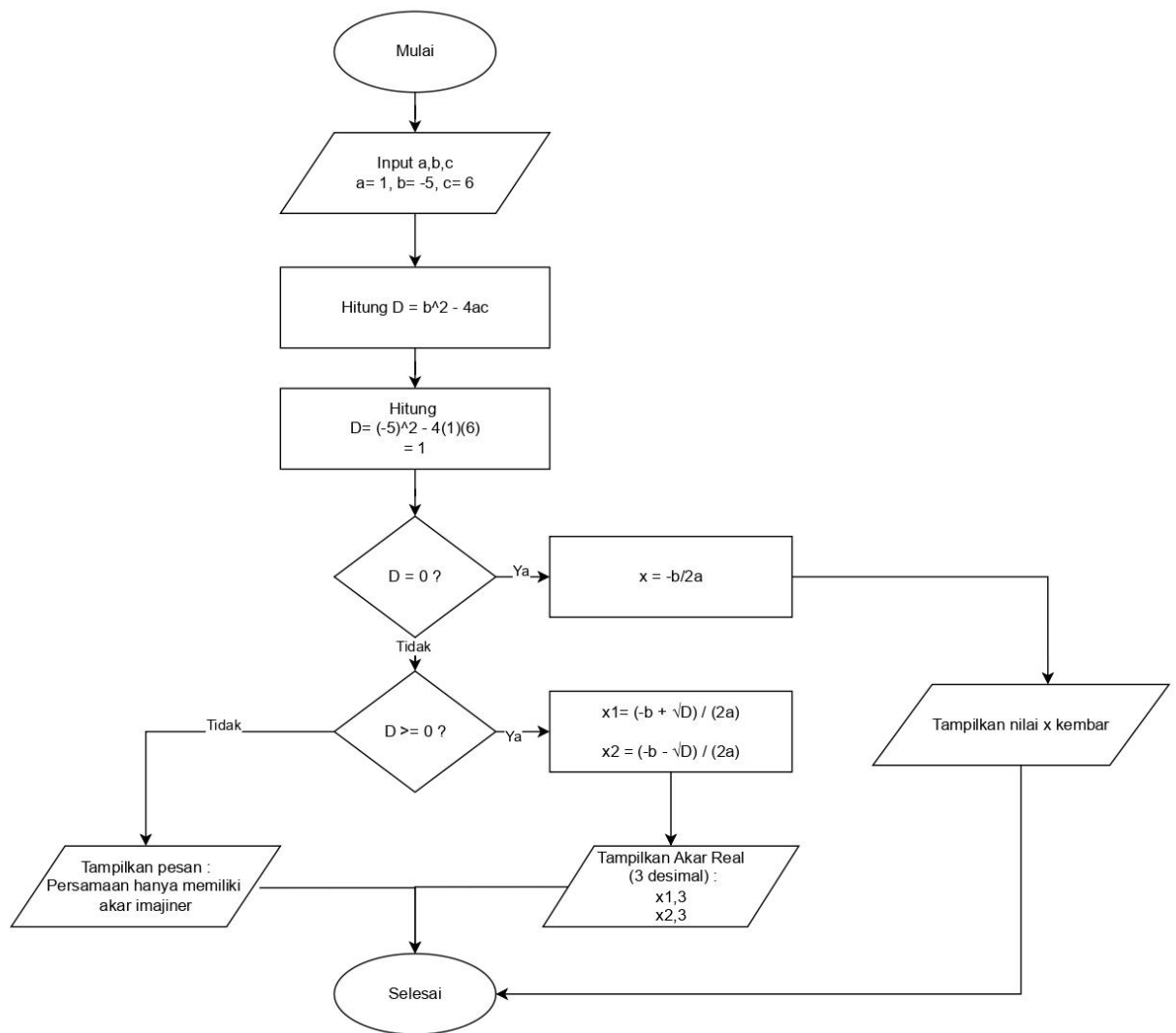
1. Program 1 :



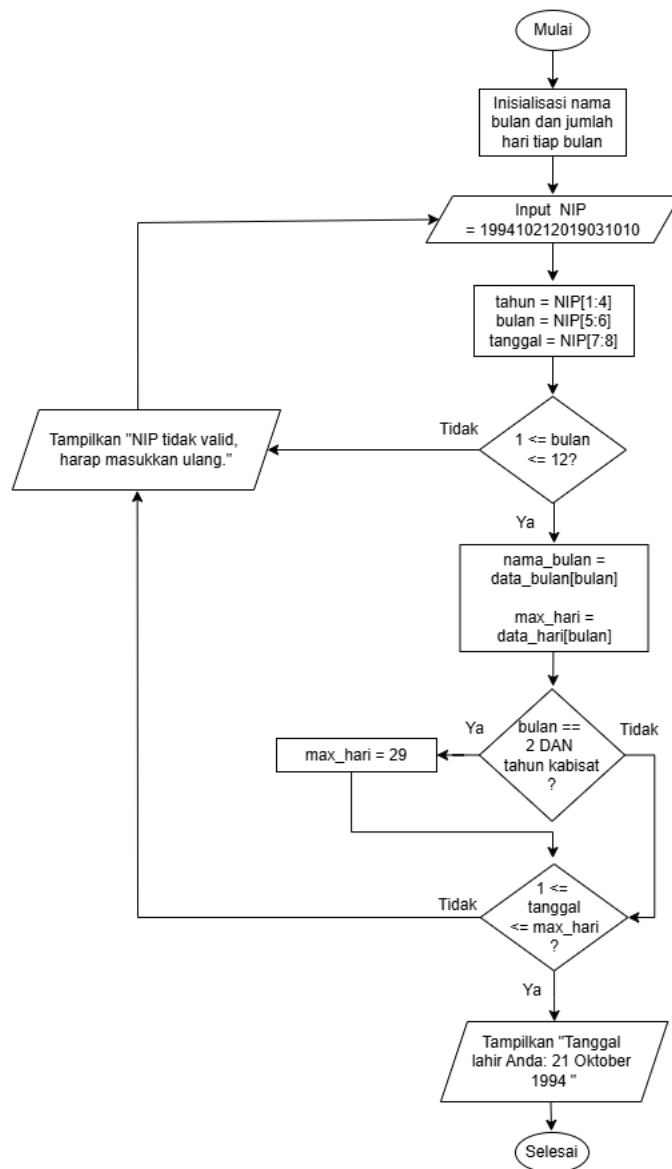
2. Program 2 :



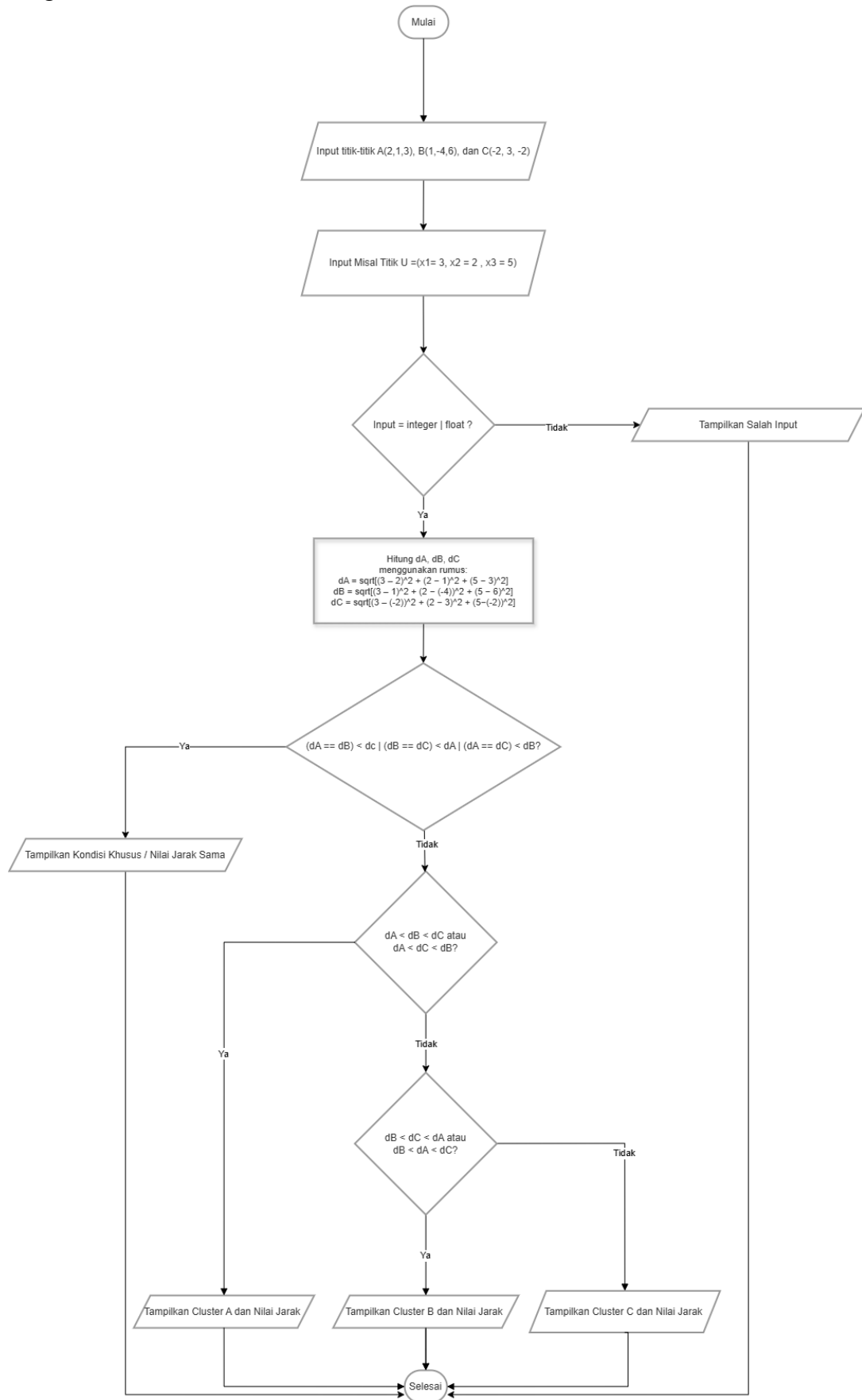
3. Program 3 :



4. Program 4 :



5. Program 5 :



6. Program 6 :

